Sahaj Bhadja

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Summary

Product-minded software engineer pursuing an MSCS at USC with 2+ years shipping React/Next.js interfaces, Python APIs, and AI-enabled data workflows. Strong in responsive UI, backend systems, relational databases, code execution workflows, and fast startup-style delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Designed and optimized SQL-backed workflow interfaces supporting 500+ users, reducing latency from **1.2s to 350ms** through query tuning, indexed access patterns, and service fixes.
- Built data validation pipelines processing 5,000+ monthly jobs with reconciliation checks, CI/CD releases, and clear runbooks, improving reliability for analytical reporting workflows.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed React and Flask product features handling 10,000+ daily requests, converting messy user workflows into responsive interfaces, REST APIs, tests, and release-ready increments.
- Built AI-assisted document grading with OCR, LLM prompts, and AWS-backed persistence, reducing feedback turnaround by **60%** while documenting edge cases and model limitations.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built Python evaluation pipelines across 100,000+ records, producing reproducible reports, debugging artifacts, and visual summaries that clarified model behavior and data quality.
- Analyzed failure patterns across generated outputs and datasets, translating complex findings into actionable recommendations while strengthening validation coverage and technical decision-making.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered API and ETL microservices using Python, AWS Glue, and S3, improving response time by **30%** while supporting scalable integrations and reliable production workflows.
- Maintained 20+ automated test suites using Jenkins, PyTest, and Mockito, increasing CI reliability by **68%**, reducing regressions, and accelerating defect isolation across services.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a MERN-based platform with WebSockets and REST APIs, supporting 2M+ synchronized records and six workflows through modular services and responsive interfaces.
- Automated Docker-based build and test flows with Jenkins, cutting release time by **50%**, improving deployment consistency, and reducing friction during validation and troubleshooting.

Skills

Frontend: React, Next.js, TypeScript, JavaScript, Tailwind CSS, responsive UI, component architecture, performance-minded UX
Backend: Python, Flask, FastAPI, Node.js, REST APIs, background jobs, auth patterns, caching concepts, service debugging
AI & Data: LLM prompts, RAG, code execution workflows, OCR, PostgreSQL, MySQL, SQL Server, MongoDB, data visualization
Delivery: Git/GitHub, Docker, AWS, GCP, CI/CD, testing, profiling basics, product collaboration, fast iteration

Projects

PiXMod Live | *Next.js, MongoDB, Stripe, Transformers*
- Built an AI-powered SaaS web app for image editing and search, integrating responsive Next.js interfaces, scalable storage, premium billing, and polished product flows.

Big-Brain Live | *LLaMA-2, FAISS, PostgreSQL, GCP*
- Built a secure document Q&A platform with semantic search, summarization, and prompt workflows over 50K+ records, reducing review time by **60%**.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Built an AI-powered SQL workspace supporting 10+ databases, combining schema inspection, query generation, Dockerized services, and debugging flows for faster data validation.